CMAI 480 Computer Vision — Course Project Proposal

Project Guidelines, Implementation Milestones, and Topic Descriptions (MATLAB Environment)

1. Project Overview

This group project provides students with practical, hands-on experience in applying computer vision techniques to solve real-world problems in surrounding communities and industry domains. Over the course of 15 weeks, student teams will design, implement, and evaluate an end-to-end computer vision system within the MATLAB software environment. The project bridges conventional computer vision algorithms with fundamental deep learning models available in MATLAB's Computer Vision and Deep Learning Toolboxes. Each team will work collaboratively in groups of 5 students, culminating in a written technical report, source code deliverables, and a live demonstration.

2. Project Requirements & Implementation Environment

- **MATLAB Environment & Toolboxes:** All primary coding, algorithm development, data analysis, and user interface design MUST be implemented using MATLAB (utilizing toolboxes such as Computer Vision Toolbox, Image Processing Toolbox, and Deep Learning Toolbox). Python or other external programming languages are strictly prohibited unless authorized for secondary hardware interface utilities.
- **Sensor & Hardware Integration:** The system must utilize a camera sensor (e.g., webcam, mobile camera, or embedded vision module) or raw image acquisition hardware connected directly to MATLAB to capture real-time images or video streams.
- **Core Topic Integration:** The project pipeline must combine at least two distinct computer vision modules (e.g., image acquisition/camera calibration paired with feature extraction, tracking, or deep learning models).
- **From-Scratch Algorithmic Step:** To ensure fundamental algorithmic understanding, each team must implement at least one core vision step (e.g., Harris corner detection, RANSAC, template matching, background subtraction, minutiae extraction, or custom filtering) from first principles/equations in MATLAB rather than relying solely on high-level library convenience functions.
- **Data Provenance & Privacy:** Projects using real-world data must source imagery from public repositories or obtain consent. Systems must provide non-biometric alternatives where applicable.
- **Project Deliverables:** A formal written technical report, fully documented MATLAB source code (.m files and/or App Designer projects), and a live system demonstration delivered during Weeks 14 and 15.

3. Suggested Project Topics by Application Domain

Smart City, Transport & Sports

- **Offside & Ball Out-of-Play Decision Support:** Pitch line detection using Hough transform/homography in MATLAB, player and ball tracking, and offside-line visualization to assist referee decision-making.
- **Runway Foreign Object Debris (FOD) Detection:** Small-object detection on road or runway surfaces using background subtraction, morphological filtering, and deep learning detectors in MATLAB.
- **Automatic License Plate Recognition (ALPR):** Plate localization, character segmentation using MATLAB morphological operations, and OCR text extraction under varying illumination.

Healthcare, Rehabilitation & Biometrics

- **Markerless Gait Symmetry Assessment:** Extracting lower-limb keypoints from side-view video, tracking joint angles (hip, knee, ankle), and measuring movement symmetry using MATLAB signal processing tools.
- **Glaucoma Optic Disc & Cup Segmentation:** Retinal image preprocessing, optic disc and cup segmentation using active contours or MATLAB U-Net architecture, and automated cup-to-disc ratio computation.
- **Fingerprint-Based Attendance & Identity Verification:** Fingerprint image enhancement via Gabor filtering, binarization, skeletonization, crossing-number minutiae extraction, and RANSAC-aligned template matching in MATLAB.

Robotics & Industrial Automation

- **Autonomous Visual Docking System:** MATLAB camera calibration, ArUco marker detection, 3D pose estimation, and visual feedback for automated docking.
- **Microsurgical Instrument Tracking:** Instrument tip segmentation, optical flow tracking, Kalman filtering during occlusion, and motion/tremor trajectory analysis in MATLAB.
- **Surface Inspection Rover:** Texture analysis, illumination correction, defect segmentation, and surface crack/corrosion identification.

Interactive Systems & AI Applications

- **Consent-Based Smart Receptionist Kiosk:** Face detection, feature extraction/embedding generation, liveness verification, and MATLAB GUI / App Designer user interface for automated appointment check-in.
- **Augmented Reality & Virtual Try-On:** Planar homography estimation, keypoint matching using MATLAB's Computer Vision Toolbox, and 3D object/graphic overlay onto live video feeds.
- **AI Video Subtitling & Text Extraction:** Text region detection, character recognition (OCR), and video frame-by-frame text overlay generation in MATLAB.

4. Project Phases and Semester Timeline

Phase	Timeline	Milestone Deliverables & Course Progression
Phase 1	**Weeks 1–3**	Form groups (5 students), select project topic, define requirements, and collect baseline dataset.
Phase 2	**Weeks 4–5**	Conduct initial literature review, image preprocessing, noise filtering, and initial feature detection/segmentation in MATLAB.
Phase 3	**Weeks 6–9**	Implement core MATLAB vision algorithms, build baseline deep learning or machine learning models, and assemble initial prototype.
Phase 4	**Weeks 10–12**	Real-time webcam/hardware integration, Camera Calibration (if needed), MATLAB App Designer GUI development, experiment execution, speed optimization, and failure-case analysis.
Phase 5	**Week 13**	Finalize technical project report, document MATLAB codebase (.m files, custom functions), and prepare submission package.
Phase 6	**Weeks 14–15**	Deliver oral presentation, conduct live MATLAB system demonstration, and undergo individual code modification and oral examination.

5. Assessment Rubric & Evaluation Scheme

Evaluation Criteria	Weight	Detailed Assessment Description
Literature Review & Problem Definition	**10%**	Comprehensive background review, ethical assessment, measurable success criteria, and clear project scope.
Methodology & MATLAB Implementation	**20%**	Correct implementation of CV/AI algorithms in MATLAB, structured code design, and inclusion of at least one from-scratch method.
Results, Analysis & System Performance	**30%**	Quantitative performance evaluation, accuracy metrics, latency/speed analysis, and robust handling of failure cases.
Technical Report & Presentation	**20%**	Professional formatting, clear data charts, well-structured written report, and effective team presentation.
Teamwork & Milestone Progression	**10%**	Equal task distribution among all 5 members, regular milestone submissions, and documented development history.
Creativity, Innovation & Oral Defense	**10%**	Novelty of solution, live MATLAB system demonstration, and individual ability to explain, reproduce, and live-modify code during the oral presentation.

6. Generative AI and Code-Generation Policy

Unless explicitly authorized for a particular activity, students may not use generative artificial intelligence or automated code-generation tools (including ChatGPT, Claude, Gemini, GitHub Copilot, or automated rewriting tools) to produce, complete, rewrite, debug, or document assessed work in CMAI 480.

Students are permitted to use standard MATLAB toolboxes (Computer Vision Toolbox, Image Processing Toolbox, Deep Learning Toolbox), authorized built-in functions, and pretrained models specifically permitted by the instructor. Use of all technical resources and external references must be fully documented and cited in the final report.

Every student must be capable of explaining, reproducing, and modifying all submitted MATLAB code. Project assessment will include supervised coding, examination of development records, individual oral questioning and live code modifications during the final demonstration. Inability to explain or reproduce submitted work will result in individual marks being withheld and referral under university academic integrity procedures.